Data Science Capstone Project
Proposal

Date:	June 29, 2026
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Proposed Title:	TsekTxt: An Explainable Web-Based Taglish Credibility Checker for Filipino Social Media Content Using Fine-Tuned Transformer Models and OCR-Enabled Screenshot Analysis
CS Research Categories/Field:	Taglish Misinformation Detection, NLP, Transformer-Based Binary Text Classification, Code-Switched Language Processing, Explainable AI, OCR / Image-to-Text, Web Application Development
UN SDG Alignment:	SDG 16: Peace, Justice, and Strong Institutions

TECHNICAL BACKGROUND:

Background of the Study

The Philippines is one of the most active social media markets in the world, with approximately 95.8 million social media users accounting for 81.9% of its total population as of late 2025 [1]. Platforms like Facebook and X (formerly Twitter) are where millions of Filipinos get their news, follow public figures, and share information daily. This same environment has made the country especially vulnerable to misinformation, which spreads quickly through informal channels before fact-checkers can respond.

A large portion of online content in the Philippines is written in Taglish, a natural blend of Filipino or Tagalog and English that Filipinos use in everyday conversation and on social media. Bautista [2] describes Taglish as a widely accepted mode of discourse among Filipinos, functioning as both a practical communication tool and a linguistic identity marker. For natural language processing (NLP) systems, however, Taglish is particularly difficult to handle because it does not follow the rules of either language consistently and is rarely found in the formal, structured datasets that most models are trained on [3].

This matters because misinformation in the Philippines does not stay in one topic area. During the 2022 presidential election, Rappler documented 256 claims from October 2021 to March 2022, of which 207 were found to be false, collectively generating over 67 million interactions on Facebook alone [4]. Arugay and Baquisal [5] further found that social media played a direct role in producing and spreading disinformation narratives that shaped voter behavior and deepened political polarization during that period. Beyond elections, the same Taglish-heavy channels carry health misinformation, scam messages, and fabricated public advisories year-round, and corrections almost always arrive too late to undo the damage.

Transformer-based models have become the leading approach for automated misinformation detection across many languages. Studies have shown that models like BERT, when fine-tuned on labeled news data, can achieve classification accuracies above 95% on standard benchmarks [6]. Multilingual variants such as mBERT and XLM-RoBERTa extend this capability across languages by training on large multilingual corpora, while language-specific models like RoBERTa-Tagalog, developed by Cruz and Cheng [7], offer stronger performance on Filipino text by training specifically on Tagalog and Filipino corpora. Each of these approaches has different strengths when applied to Taglish, and no existing tool has made any of them available to general Filipino users through an accessible web interface.

Existing misinformation detection tools do not address this gap. BRENDA [8] and FakeZero [9] both showed that transformer models can work in real-time browser environments, while SEMiNExt [10] demonstrated point-of-access intervention during web searches. All three, however, were built and evaluated on English content and do not account for the structural and linguistic characteristics of Taglish. No publicly accessible tool currently exists that can assess the credibility of informal Filipino social media text.

TsekTxt addresses this directly. It is a web application where Filipino users can paste any Taglish text or upload a screenshot and receive a credibility assessment in real time, classified as suspicious, or not suspicious. The classifier is fine-tuned on the Fake News Filipino Dataset [11] and the Philippine Fake News Corpus [12] as its initial training domain, using election misinformation data as a starting point while being designed to handle the broader range of Taglish content that Filipinos encounter daily. The goal is to give ordinary users, educators, journalists, and fact-checkers a practical and language-appropriate tool they can actually use.

STATEMENT OF THE PROBLEM:

TsekTxt aims to address the absence of an accessible, web-based credibility checking tool designed specifically for informal Taglish social media content. The study seeks to address the following specific problems:

1. The lack of a fine-tuned binary credibility classifier for Taglish social media text — specifically, fine-tuning XLM-RoBERTa on a combined Filipino misinformation corpus (Fake News Filipino Dataset and Philippine Fake News Corpus) of approximately 26,000 labeled samples for binary classification of Taglish content as Suspicious or Not Suspicious, with Philippine election misinformation serving as the initial classification domain.
2. Development of a web application (TsekTxt) that can:
    - Accept Taglish text typed or pasted directly by the user, or extract text from an uploaded screenshot or image through the Groq API's multimodal vision capabilities, to accommodate how misinformation actually circulates among Filipino users
    - Classify submitted content in real time as Suspicious or Not Suspicious, accompanied by a confidence score
    - Highlight the specific words or tokens most strongly influencing the classification decision to support the user's own critical assessment before believing or sharing content
    - Support future domain expansion beyond election misinformation to cover adjacent Taglish-heavy threat categories, including health misinformation, scam messages, fabricated disaster alerts, and misleading public advisories
3. To evaluate the developed web application against ISO 25010 standards using the Functionality, Usability, Reliability, Performance, and Security (FURPS) model

HOW DID OTHERS SOLVE THE PROBLEM?

Botnevik, Sakariassen, and Setty [8] proposed BRENDA, a browser extension designed to automate the credibility assessment of false claims encountered on the web. The fact checker utilizes a deep neural network-based architecture for recognizing fact-check worthy claims and classifying them on-the-fly while returning results with corresponding evidence embedded into the browser window. The problem they aimed at was the massive proliferation of online misinformation, making traditional manual fact-checking impossible to be performed by fact-checking platforms such as Snopes and Politifact. The architecture of their model is client-server based where the content script retrieves the text or the entire URL of the current website and sends it to a RESTful API running on the backend server. Results are returned and displayed in a popup within the extension interface. Through BRENDA, it is possible to integrate the fake news detection technique based on the transformer model into the regular process of web browsing without forcing the user to leave the current page. Nevertheless, BRENDA works only with web articles written in English language and cannot handle code-switching or social media posts, let alone annotate them inline.

Essahli, Sarsar, Fouad, and Bentajer [9] presented FakeZero, a fully client-side browser extension that flags unreliable posts on Facebook and X in real time while the user scrolls. Unlike server-dependent approaches, all computation including DOM scraping, tokenization, transformer inference, and interface rendering runs locally on the user's device through the Chromium messaging API, ensuring that no personal data leaves the device. They focused on tackling the two-pronged challenge of high accuracy rates in detecting misinformation as well as user privacy, as all the previous approaches had been based on one of two factors: inference done by server systems or the use of a static, transformer-less architecture incapable of understanding semantics. They used three distinct stages of training including baseline fine tuning, domain adaptation using focal loss, adversarial augmentations, and post-training quantization. Their DistilBERT-based quantized model achieved 97.1% macro-F1 and a median inference latency of approximately 103 milliseconds on commodity hardware. FakeZero demonstrated that high-accuracy transformer inference is feasible entirely within the browser sandbox. Nonetheless, the model has been trained and tested solely based on English-based posts and lacks consideration of the language properties in Taglish and other code-switched languages as well as lack of pre-training in languages from Southeast Asia.

Ismail et al. [10] proposed an extension for the web search engine named SEMiNExt, which is meant to alert users to any web pages that might have information that could mislead them. Their aim was to tackle the problem of unintentional encounters with misinformation when users search the web, as they believed that interventions at points of contact would prove more effective than any remedial action later on. The technology does not involve the classification of social media posts but focuses on the analysis of content on web pages. SEMiNExt demonstrated the viability of browser-level misinformation intervention and provided evidence that NLP-backed classifiers can operate effectively within a browser extension context with a comparably scoped evaluation set. However, the system targets general English-language web content and does not engage with the specific challenges of informal, code-switched Filipino social media text.

Szczepanski, Pawlicki, Kozik, and colleagues [13] introduced an explainability method for BERT-based fake news detection models, applying attention-based attribution to identify which tokens most strongly influenced a model's credibility classification. They addressed the opacity of transformer-based classifiers, which tend to operate as black boxes despite their strong performance, limiting user trust and interpretability. The authors have made a visualization of the token-wise attention weights that allows revealing the linguistic attributes responsible for making predictions by the models. It was shown in their paper that adding interpretability components to fake news detection algorithms enhances transparency while minimizing adverse effects on classification quality. While their work focused on English-language news articles rather than social media content, their findings on attribution-based explainability inform the interpretability component of transformer-based misinformation detection systems more broadly.

Filipino fake news detection. Cruz, Tan, and Cheng [11] created the Fake News Filipino dataset and explored transfer learning approaches for low-resource Filipino text classification. Their study demonstrated that the limitation of large annotated datasets can be overcome by training machine learning models to classify Filipino fake news. The work presented here is highly relevant to the proposed Taglish Credibility Checker since the system will also use Filipino misinformation data as the basis for training a credibility classification model. However, the study was mainly aimed at creating datasets and benchmarking models, rather than developing a web-based credibility checker that would be accessible to the average user for assessing questionable online content.

Cruz and Cheng [7] advanced natural language processing for Filipino by improving large-scale Filipino language resources and developing RoBERTa-based pretrained models for Filipino text. Their work shows that building Filipino-specific transformer models can help improve performance on downstream Filipino NLP tasks. This is relevant to the proposed system because Taglish credibility detection can benefit from language models that better understand the Filipino vocabulary, grammar and local linguistic patterns. The study aims to improve Filipino NLP resources in general while the proposed capstone will implement Filipino NLP models to a practical web-based misinformation credibility checker.

ClaimBuster is an automated fact-checking system introduced by Hassan, Arslan, Li and Tremayne [14], which uses natural language processing and supervised learning to identify factual claims that are check-worthy. The algorithm is used to reduce the load on human fact-checkers by ranking the statements that need verification. The algorithm is very necessary in the proposed system, since in detecting misinformation, the system must not only classify the text but also assist users to verify suspicious contents. ClaimBuster was built primarily for English political discourse and is not suitable for Filipino, Taglish, or code-switched online material.

Full Fact [15] has created AI-assisted fact-checking tools that help fact-checkers to more effectively track, find and counter false or misleading claims. Their system demonstrates that artificial intelligence can be an aid to fact-checking groups, rather than a wholesale replacement for human judgment. This is important to the proposed Taglish Credibility Checker because the system should not be presented as an official fact-checking body. Rather, it serves as a credibility aid, providing users with a classification outcome, a confidence level and supporting cues to foster critical assessment prior to believing or sharing online material.

HOW DO YOU PLAN TO SOLVE THE PROBLEM?

The proponents of this study will develop TsekTxt using XLM-RoBERTa as its core classification engine. XLM-RoBERTa was selected for its strong cross-lingual performance on low-resource and code-switched language tasks, making it well-suited for the informal, mixed-language nature of Taglish social media content. The model will be fine-tuned on a combined Filipino misinformation corpus consisting of the Fake News Filipino Dataset [11] and the Philippine Fake News Corpus [12], which together yield approximately 26,000 labeled samples for binary credibility classification (Suspicious / Not Suspicious). Fine-tuning will be conducted on Google Colab using a GPU runtime environment, using Python and the HuggingFace Transformers library. The resulting model checkpoint will be saved and loaded as the production classifier, served through a REST API built with FastAPI.

The web application frontend will be built using React.js and will support two input pathways: direct text entry and image or screenshot upload. For image-based submissions, the system will route the uploaded file through the Groq API, which will leverage its multimodal vision capabilities via LLaMA 3.2 Vision to extract Taglish text from screenshots to extract Taglish text from screenshots before passing the output to the classifier. This approach is more effective than traditional OCR for informal Filipino social media content, as Groq can handle code-switched text, stylized fonts, meme formats, and inconsistent layouts that standard character recognition tools frequently misread. Every classification response will include a Suspicious or Not Suspicious label, a confidence percentage, and a token-level attribution display generated through Integrated Gradients to surface the specific words driving the prediction in plain, non-technical terms. The system's initial domain is Philippine election misinformation, and its architecture is designed to accommodate domain expansion through additional fine-tuning datasets in future iterations covering health misinformation, scam messages, and disaster-related content. The completed application will be evaluated against ISO 25010 standards using the FURPS model through structured user acceptance testing with students, educators, journalists, and fact-checkers as target participants.

REFERENCES:

[1] S. Kemp, “Digital 2026: The Philippines,” DataReportal, 2025. https://datareportal.com/reports/digital-2026-philippines

[2] M. L. S. Bautista, “Tagalog-English code switching as a mode of discourse,” Philippine Journal of Linguistics, vol. 35, no. 1, pp. 25-38, 2004. https://files.eric.ed.gov/fulltext/EJ720543.pdf

[3] M. Herrera, A. Aich, and N. Parde, “A dataset for investigating Tagalog-English code-switching,” in Proc. LREC, 2022, pp. 2090-2097. https://aclanthology.org/2022.lrec-1.225.pdf

[4] Rappler, “Fake news and internet propaganda, and the Philippine elections: 2022,” May 2022. https://www.rappler.com/technology/features/analysis-fake-news-internet-propaganda-2022-philippine-elections/

[5] A. A. Arugay and J. K. A. Baquisal, “Mobilized and polarized: Social media and disinformation narratives in the 2022 Philippine elections,” Pacific Affairs, vol. 95, no. 3, pp. 549-573, 2022.

[6] A. Alghamdi, “Enhancing fake news detection with transformer-based deep learning: A multidisciplinary approach,” PLOS ONE, 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12419611/

[7] J. C. B. Cruz and C. Cheng, “Improving large-scale language models and resources for Filipino,” in Proc. LREC, 2022, pp. 6548-6555.

[8] B. Botnevik, E. Sakariassen, and V. Setty, “BRENDA: Browser extension for fake news detection,” in Proc. 43rd ACM SIGIR, 2020, pp. 2117-2120.

[9] S. Essahli, O. Sarsar, I. Fouad, A. Bentajer, and A. Motii, “FakeZero: Real-time, privacy-preserving misinformation detection for Facebook and X,” arXiv, 2025. https://arxiv.org/abs/2510.25932

[10] S. Ismail, T. N. M. Aris, A. Suliman, and M. Ahmad, “SEMiNExt: Web search engine misinformation notifier extension,” Healthcare, vol. 9, no. 2, Art. 218, 2021. https://doi.org/10.3390/healthcare9020218

[11] J. C. B. Cruz, J. A. Tan, and C. Cheng, “Localization of fake news detection via multitask transfer learning,” in Proc. LREC, 2020, pp. 2596-2604.

[12] A. C. T. Fernandez, “Philippine Fake News Corpus,” GitHub, 2019. https://github.com/aaroncarlfernandez/Philippine-Fake-News-Corpus

[13] M. Szczepanski, M. Pawlicki, R. Kozik, and M. Choraś, “New explainability method for BERT-based model in fake news detection,” Scientific Reports, vol. 11, Art. 23705, 2021. https://doi.org/10.1038/s41598-021-03100-6

[14] N. Hassan, F. Arslan, C. Li, and M. Tremayne, “Toward automated fact-checking: Detecting check-worthy factual claims by ClaimBuster,” in Proc. 23rd ACM SIGKDD International Conference on

Knowledge Discovery and Data Mining, 2017, pp. 1803–1812.
https://doi.org/10.1145/3097983.3098131

[15] Full Fact, "Full Fact AI," Full Fact. https://fullfact.org/ai/

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